The Traveling Microscope and Newton’s Rings

Interference of equal thickness is caused by non-parallel films. Newton's rings and wedge interference are typical interference of equal thickness. Interference of equal thickness can be used to measure the curvature radius, filament diameter, and flatness of plane.

The traveling microscope is an instrument well-suited for the purpose of measuring small distances with high accuracy. The curvature radius of Newton's rings glass can be calculated by measuring the diameter of Newton's rings.

Objectives

1. Learn the principle of Newton's rings interference.
2. Learn the structure, optical principle, and usage of a traveling microscope.
3. Learn how to measure the curvature radius of Newton's rings glass by using a traveling microscope.

Apparatus

JCD3 traveling microscope, Newton's rings glasses, Sodium vapour lamp.

Theory

The Newton's rings glasses consist of a convex lens with a large curvature radius stacked on a flat glass, forming a thin film of air with increasing thickness from the centre outwards. When parallel monochromatic light is vertically irradiated onto the Newton's rings glasses from top to bottom, two beams of light reflected from the upper and lower interfaces of the air film meet at the convex surface of the lens, causing interference. At the same distance from the centre, the thickness of the air film is equal, so that the optical path difference between the two beams of light reflected is equal. Newton's rings are a group of concentric rings centred around the contact point.

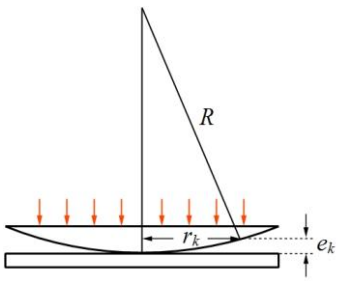

Fig. 1 The structure of Newton's rings glasses

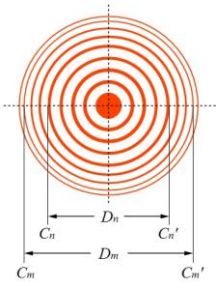

Fig. 2 Newton's rings

The geometric path difference between the two reflected lights is twice the thickness of the air film, and add half wave loss, the optical path difference δ is

$$\delta = 2e + \frac{\lambda}{2} \tag{1}$$

When the optical path difference is an odd multiple of the half wavelength, dark stripes are formed.

$$\delta = 2e_k + \frac{\lambda}{2} = (2k+1)\frac{\lambda}{2}, \quad k = 0,1,2,3\ldots \tag{2}$$

When the optical path difference is an even multiple of the half wavelength, bright stripes are formed.

$$\delta = 2e_k + \frac{\lambda}{2} = k\lambda, \quad k = 0,1,2,3\ldots \tag{3}$$

The relationship between the radius of the k-th dark ring r_k and the thickness of the air film e_k is:

$$r_k{}^2 = R^2 - (R - e_k)^2 = 2R \cdot e_k - e_k{}^2 \tag{4}$$

When the curvature radius of the lens R is much greater than e_k, e_k is negligible.

$$e_k = \frac{r_k{}^2}{2R} \tag{5}$$

We can get

$$r_k{}^2 = kR\lambda, \quad k = 0,1,2,3\ldots \tag{6}$$

If the wavelength of monochromatic light is known, the curvature radius of the lens R can be calculated by measuring the radius of the kth dark ring r_k.

In practice, there is not an ideal point contact between two pieces of glass, so the interference pattern presented at the centre point is not a dark spot, but a fuzzy circular dark spot. This makes it difficult to determine the number k of a certain circle of dark stripes. In fact, it is difficult to calculate the curvature radius R of the lens using the above formula.

In practical measurements, dark rings are relatively easy to observe, so we select the mth and n^{th} dark rings and measure their diameters to calculate the radius of curvature R of the lens.

$$r_m^{\ 2} = mR\lambda \tag{7}$$

$$r_n^{\ 2} = nR\lambda \tag{8}$$

$$r_m^{\ 2} - r_n^{\ 2} = (m-n)R\lambda \tag{9}$$

$$R = \frac{r_m^{\ 2} - r_n^{\ 2}}{(m-n)\lambda} = \frac{D_m^{\ 2} - D_n^{\ 2}}{4(m-n)\lambda} \tag{10}$$

Contents

The traveling microscope mainly consists of an eyepiece, an objective lens, a handwheel, a mirror, and a reading device.

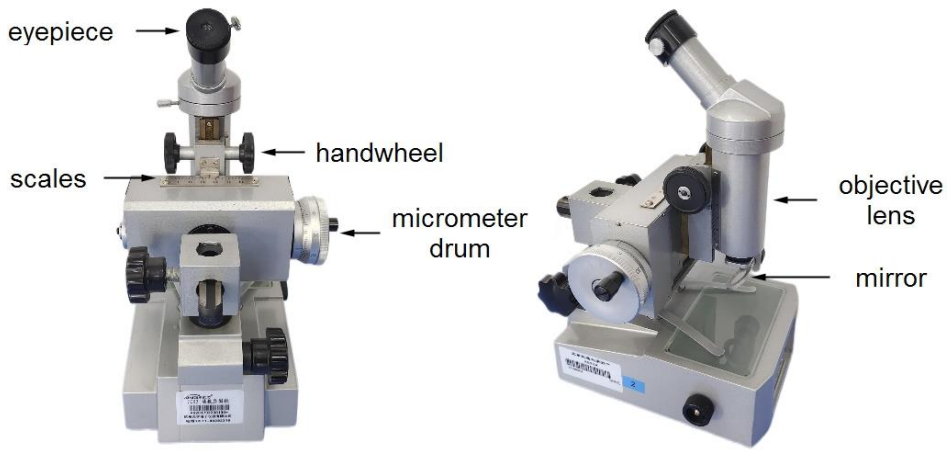

Fig. 3 The structure of the traveling microscope

Newton's rings glasses consist of two pieces of glass and three screws.

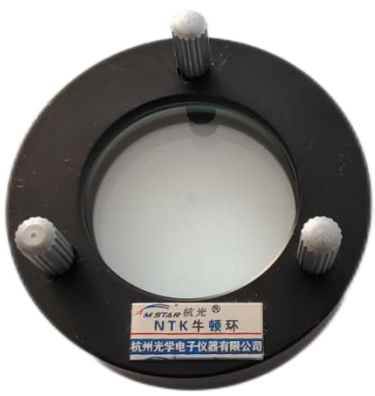

Fig. 4 Newton's rings glasses

1. Adjustment of eyepiece

Turn on the Sodium vapour lamp. Adjust the eyepiece focus knob until the cross-wire in eyepiece view is clear. Adjust the direction of the cross-wire to make them to be horizontal and vertical.

2. Adjustment of mirror

Place the Newton's rings glasses under the objective lens, and align its central contact point with the objective lens. Adjust the mirror so that it is approximately 45° as shown in Fig.5. Then fine-tune the knob of the mirror until the light in eyepiece view is the brightest.

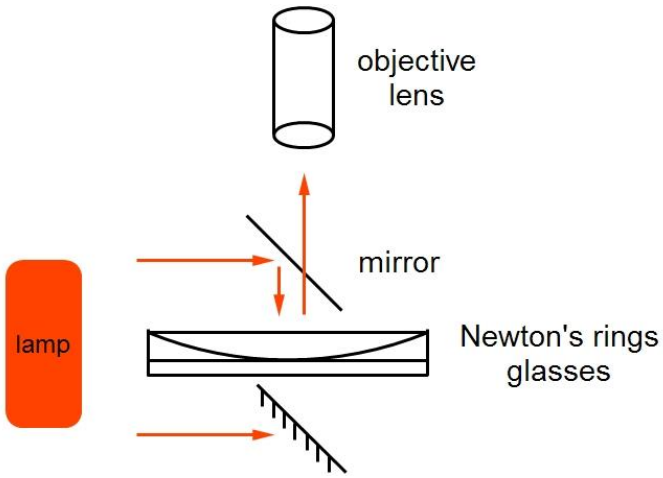

Fig. 5 Optical principle of measurement

3. Adjustment of objective lens

Rotate the handwheel slowly to move the objective lens from bottom to top until the Newton's rings are observed clearly in eyepiece view. Move the Newton's rings glasses to make the centre of the Newton's rings coincides with the centre of the cross-wire in eyepiece view.

4. Record data

The reading device consists of a main scale and a micrometer drum. There are 0-50mm lines on the scale, with each grid of 1mm; the micrometer drum rotates for one full revolution (100 cells), and the corresponding scale moves by 1mm. Therefore, each small cell of the micrometer drum represents 0.01 mm. The third digit after the decimal point is estimated.

5. Measure the curvature radius of Newton's rings glass

Take $m = 15$, $n = 5$, rotate the micrometer drum to move the cross-wire horizontally from the centre of Newton's rings to several rings outside the 15th ring, and then reverse it back to the 15th ring. When the vertical line of the cross-wire coincides with the 15th ring, read data C_{15}, move the cross-wire in the same direction

to the 5th ring and read C_5. Continue moving the cross-wire in the same direction, read C_5' and C_{15}' on the other side.

Rotate the Newton's rings glasses, repeat the above measurement 6 times, and record them in Table 1.

Calculate the curvature radius of Newton's rings glass by Eq. (10). The wavelength of Sodium light is 589.3nm.

6. Measurement error

During the measurement, it is necessary to pay attention to the impact of return error. The micrometer drum should be rotated in one direction. It is prohibited to reverse halfway and then move forward.

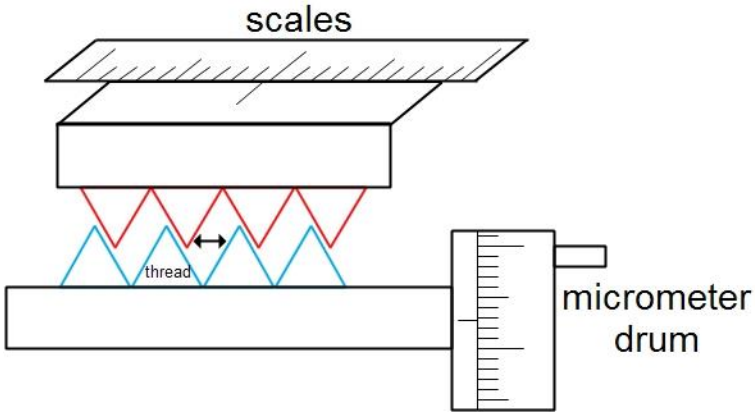

Fig. 6 The return error

Precautions

1. Keep the Sodium vapour lamp close to the traveling microscope to ensure sufficient light intensity.

2. The reflective mirror under the base of the traveling microscope does not need to be used, and its reflective surface should be turned inward.

Data Taking

Table 1 Measurement of Newton's rings

No.	Measured Data(mm)				Diameter(mm)		Curvature Radius(mm)
	C_{15}	C_5	C'_5	C'_{15}	D_{15}	D_5	R
1							
2							
3							
4							
5							
6							

Data Analysis

Calculate the uncertainty of the curvature radius of Newton's rings glass. The calculation steps are as follows:

Calculate the mean value of D_{15} and D_5 .

Calculate the uncertainty of D. D_5 and D_{15} are measured by the same instrument, so their uncertainty is the same.

$$u(D) = u(D_{15}) = u(D_5)$$

Calculate the standard deviation of the mean value as the class A component of the uncertainty:

$$u_A = S_{\overline{D_{15}}} = \frac{1}{\sqrt{n}}\sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(D_i - \overline{D_{15}})^2}$$

Calculate the class B component of the uncertainty (Δ_m =0.005mm):

$$u_B = \frac{1}{\sqrt{3}}\Delta_m$$

Calculate the standard uncertainty of D_{15} (or D_5) :

$$u(D_{15}) = \sqrt{u_A^2 + u_B^2}$$

R is the indirect calculation quantity, calculate the relative uncertainty of R:

$$E(R) = 2\frac{\sqrt{\overline{D_{15}}^2 + \overline{D_5}^2}}{\overline{D_{15}}^2 - \overline{D_5}^2} u(D) \times 100\%$$

Calculate the mean value of *R*.

Calculate the standard uncertainty of *R*.

$$u(R) = \overline{R} \cdot E(R)$$

$$R = \overline{R} \pm u(R)$$

Questions

1. When measuring the diameter of a Newton's ring, if the cross-wire missed the position needed to stop for reading, can you reverse the micrometer drum back to that position and take a reading? Why?